Sales Performance Analysis Using SQL and Power BI

◆ 1. Project Overview

This project focuses on analyzing sales data to uncover key insights related to revenue, profitability, and customer behavior. The objective is to build an end-to-end data analytics solution involving data cleaning, database management, SQL analysis, and dashboard visualization.

◆ 2. Objective

The main objectives of this project are:

- To clean and preprocess raw sales data
- To store and manage data using PostgreSQL
- To perform SQL-based analysis for business insights
- To build an interactive dashboard using Power BI
- To identify trends, patterns, and key performance indicators

◆ 3. Tools and Technologies Used

- **Python** – Data cleaning and preprocessing
- **PostgreSQL** – Data storage and querying
- **Power BI** – Data visualization and dashboard creation

◆ 4. Dataset Description

The dataset contains sales transaction data including:

- Order details (Order ID, Order Date, Ship Date)
- Customer information (Customer Name, Segment, Region)
- Product details (Category, Sub-category, Product Name)
- Sales metrics (Sales, Quantity, Discount, Profit)

This dataset enables analysis of sales performance across different dimensions such as time, geography, and product categories.

5. Data Cleaning and Preparation

Data cleaning was performed using Python to ensure data quality and consistency. The following steps were carried out:

- Loading the data

```
[3]: #loading the data
     import pandas as pd
     df = pd.read_csv("superstore_analysis.csv", encoding="ISO-8859-1")
     df.drop(columns=['Column1'], inplace=True)
     print(df.head())
```

```
         Order ID  Order Date   Ship Date       Ship Mode Customer ID  \
0  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CG-12520
1  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CG-12520
2  CA-2016-138688   6/12/2016   6/16/2016    Second Class    DV-13045
3  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO-20335
4  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO-20335

     Customer Name    Segment        Country             City       State  \
0      Claire Gute   Consumer  United States        Henderson    Kentucky
1      Claire Gute   Consumer  United States        Henderson    Kentucky
2  Darrin Van Huff  Corporate  United States      Los Angeles  California
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale     Florida
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale     Florida

   Postal Code Region       Product ID         Category Sub-Category  \
0        42420  South  FUR-BO-10001798        Furniture    Bookcases
1        42420  South  FUR-CH-10000454        Furniture       Chairs
2        90036   West  OFF-LA-10000240  Office Supplies       Labels
```

- Handling missing values, Removing duplicates

```
[11]: #checking if their is any duplicate value in dataset
      df.duplicated().any()
[11]: True
[12]: df.duplicated().sum()
[12]: 1
[13]: df[df.duplicated(keep=False)]
```

[13]:

	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	city	state	postal_code	region	product_id	category	sul catego
3405	US-2014-150119	2014-04-23	2014-04-27	Standard Class	LB-16795	Laurel Beltran	Home Office	United States	Columbus	Ohio	43229	East	FUR-CH-10002965	Furniture	Chai
3406	US-2014-150119	2014-04-23	2014-04-27	Standard Class	LB-16795	Laurel Beltran	Home Office	United States	Columbus	Ohio	43229	East	FUR-CH-10002965	Furniture	Chai

```
[14]: #dropping the duplicate item
      df = df.drop_duplicates()
[15]: df.duplicated().any()
[15]: False
```

- Converting data types (dates, numeric values)

- Creating new features such as:
  - Year and Month from order date
  - Profit Margin
- Detecting outliers in sales data

This step ensured that the dataset was ready for analysis.

🔹 6. Data Storage (PostgreSQL)

The cleaned dataset was loaded into a PostgreSQL database for efficient querying and analysis.

- A database was created to store the sales data
- Tables were defined with appropriate data types
- Data was inserted using Python (SQLAlchemy)
- SQL queries were executed to analyze the data

🔹 7. SQL Analysis

SQL was used to extract meaningful insights from the data. Key analyses include:

- Total Sales and Profit calculation
- Monthly sales trends
- Sales by category and region
- Identification of top-performing products
- Detection of loss-making products
- Analysis of repeat customers
- Impact of discount on profitability

```sql
--1.calculating the total sales and profit
select
SUM(sales) as total_sales,
SUM(profit) as total_profit
from sales;

--2. Calculating the monthly sales trend
select
year,
month,
sum(sales) as total_sales
from sales
group by year, month
order by year, month;

--3. Calculating the sales by category
select
category,
sum(sales) as total_sales
from sales
group by category
order by total_sales desc;

--4. Profit by Region
select
regio,
sum(profit) as total_profit
from sales
group by country
order by total_profit desc;
```

These queries helped uncover patterns and support decision-making.

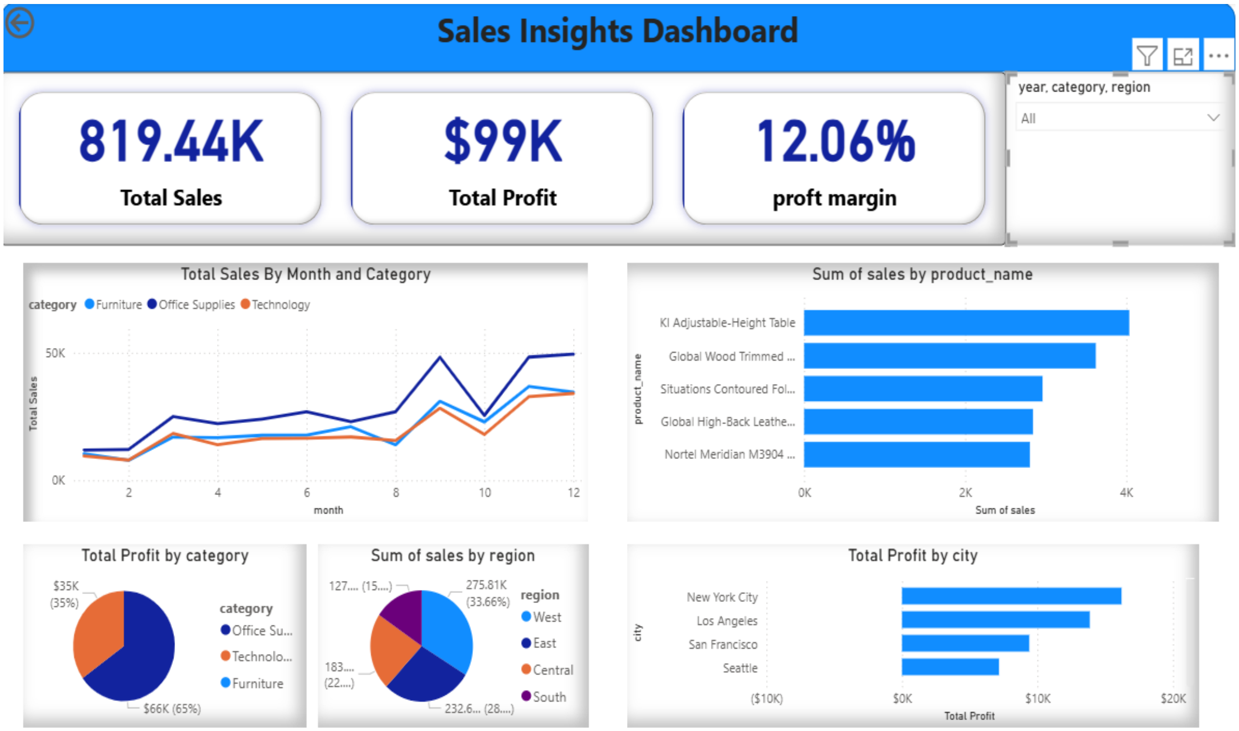
- **8. Dashboard Development (Power BI)**

An interactive dashboard was created using Power BI to visualize key insights.

Key Features:

- **KPI Cards**
    - Total Sales
    - Total Profit
    - Profit Margin
- **Visualizations**
    - Sales trend over time (line chart)
    - Sales by category (bar chart)
    - Profit by region
    - Top 10 products by sales
    - Repeat vs one-time customers
    - Discount vs profit analysis
- **Filters (Slicers)**
    - Year
    - Category
    - Region

The dashboard enables users to interactively explore the data and gain insights.



◆ 9. Key Insights

- Sales show a strong upward trend with peaks during November and December
- The Technology category generates the highest revenue
- High discount levels negatively impact profitability
- A significant portion of revenue comes from repeat customers
- Some products generate high sales but result in low or negative profit

◆ 10. Conclusion

This project demonstrates the complete data analysis workflow, from data cleaning to visualization. By leveraging SQL and Power BI, meaningful insights were extracted to support business decisions.

The project highlights the importance of data-driven strategies in improving sales performance and profitability.

◆ 11. Future Scope

- Integration with real-time data sources
- Advanced analytics using machine learning
- Deployment of dashboard for business use